Regional Salmon Abundance

(1) Metadata Reference Information

Metadata Last Updated: 20251112

Dataset URL: https://data.salmonwatersheds.ca/result?datasetid=551

Contact:
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(2) Identification Information

Data Identification:

Supplemental Information

-989898 is the PSF database code for a null value.

Attributes:

region	One of ten regions defined for organizing data in the PSE: Yukon, Northern Transboundary, Haida Gwaii, Nass, Skeena, Central Coast, Fraser, East Vancouver Island & Mainland Inlets, West Vancouver Island, Columbia
species	One of six species: Chinook, chum, coho, pink, sockeye, steelhead
year	Calendar year the estimates correspond to (i.e., return year)
spawners	The raw spawner abundance estimate (may be an index, model estimate, or absolute count depending on region and species)
smoothedSpawners	Spawners smoothed using a right-aligned running geometric mean over the dominant generation length for the region and species.
spawnersAnomaly	The smoothed spawners, transformed to be a per cent anomaly from the long-term average. The most recent spawnersAnomaly is the current state of the regional salmon population.
runsize	The raw run size estimate (a.k.a. total abundance; may be an index, model estimate, or absolute count depending on region and species)
smoothedRunsize	Run size smoothed using a right-aligned running geometric mean over the dominant generation length for the region and species.
runsizeAnomaly	The smoothed run size, transformed to be a per cent anomaly from the long-term average. The most recent runsizeAnomaly is the current state of the regional salmon population.

spawners_short_trend	The predicted number of spawners from a linear trend fitted over the most recent three generations.
spawners_short_trend_lwr	The lower 95% prediction interval on the number of spawners from a linear trend fitted over the most recent three generations.
spawners_short_trend_upr	The upper 95% prediction interval on the number of spawners from a linear trend fitted over the most recent three generations.
spawners_long_trend	The predicted number of spawners from a linear trend fitted over the entire time series.
spawners_long_trend_lwr	The lower 95% prediction interval on the number of spawners from a linear trend fitted over the entire time series.
spawners_long_trend_upr	The upper 95% prediction interval on the number of spawners from a linear trend fitted over the entire time series.
runsize_short_trend	The predicted run size from a linear trend fitted over the most recent three generations.
runsize_short_trend_lwr	The lower 95% prediction interval on the run size from a linear trend fitted over the most recent three generations.
runsize_short_trend_upr	The upper 95% prediction interval on the run size from a linear trend fitted over the most recent three generations.
runsize_long_trend	The predicted run size from a linear trend fitted over the entire time series.
runsize_long_trend_lwr	The lower 95% prediction interval on the run size from a linear trend fitted over the entire time series.
runsize_long_trend_upr	The upper 95% prediction interval on the run size from a linear trend fitted over the entire time series.
source_id	Unique identifier that describes the source of the raw data used for spawners and run size. See below for full references associated with each source_id.
datasetversion	Date the dataset was last updated in PSF's database.

Extent
Description_of_Geographic_Extent: British Columbia and Yukon

Temporal extent:
Beginning_Date: 1893-01-01
Ending_Date: 2024-12-31

Status: Complete.

Maintenance and Update Frequency: As needed.

Citation:

Title: Regional Salmon Abundance

Publication Date: 20251112

Responsible Party: Pacific Salmon Foundation
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Description:

Abstract: This dataset includes time series of aggregate raw and smoothed spawner abundance and total abundance (a.k.a. run size) for six species (Chinook, chum, coho, pink, sockeye, steelhead) in each of ten regions.

Purpose: This dataset was developed by the Pacific Salmon Foundation's Salmon Watersheds Program as part of PSF's State of Salmon Report. The State of Salmon Report is a broad-scale evaluation of the state of all six species of Pacific salmon found in British Columbia and the Yukon, Canada. This flagship publication from the Pacific Salmon Foundation takes a data-driven approach to summarizing the status and trends in regional abundance for each species of salmon in ten major salmon-bearing regions in western Canada.

Resource Constraints:

Data Set Credit:

This dataset is based on core data from a variety of sources including the Pacific Salmon Commission and DFO. Full references associated with the dataset source_id field are provided in the table below. Further details are provided in the State of Salmon Report technical documentation available at https://salmonwatersheds.github.io/state-of-salmon/. The methods and models used to produce this dataset are based on published literature, agency reports, and knowledge of experts working with the species.

source_id	*Reference*
JTC_202503	Yukon River Joint Technical Committee. 2025. Yukon River Salmon 2024 Season Summary and 2025 Season Outlook. Yukon River Joint Technical Committee Report YUKON JTC (25)-01: vii + 186 p. Available from https://www.yukonriverpanel.com/publications/yukon-river-joint-technical-committee-reports/.
PSC_20250416	Chinook Technical Committee Annual Report Data App. Release 2025.01.v1. Pacific Salmon Commission: Chinook Technical Committee. [Accessed 2025 Apr 16]. https://psc1.shinyapps.io/CTC_annual_report_data/_w_637af0d3/_w_8625d32f/
Foos_20250415	Foos, Aaron. 2025. Preliminary Estimates of Transboundary River Salmon Production, Harvest and Escapement compiled for the Pacific Salmon Commission Transboundary Technical Committee. Provided to the Pacific Salmon Foundation on April 15, 2024.
NuSEDS_20250221	DFO. 2025. The New Salmon Escapement Database System (NuSEDS). Available from Open Data Canada at https://open.canada.ca/data/en/dataset/c48669a3-045b-400d-b730-48aafe8c5ee6 [accessed 2025-02-22].
CTC_20250714	Chinook Technical Committee. 2025. Annual Report of Catch and Escapement for 2024. Pacific Salmon Commission Joint Chinook Technical Committee Report **TCCHINOOK (25)-02**: xviii + 180 p. Available from https://www.psc.org/reports/tcchinook-25-02.

English_20230930	English, K.K., Alexander, R.F., Ian A. Beveridge, Challenger, W., Percival, N., Hertz, E., and Atkinson, C. 2023. Preliminary Area 3 Salmon (2018–2022) and Nass River Summer-Run Steelhead (1994–2022) Escapement, Catch, Run Size, and Exploitation Rate Estimates. Prepared by LGL Ltd. and Nisga'a Lisims Government - Fisheries & Wildlife Department for the Pacific Salmon Foundation and Nisga'a-Canada-BC Treaty's Joint Fisheries Management Committee. Available from https://salmonwatersheds.ca/document/lib_526/.
Nisgaa_20241203	Nisga'a Fisheries and Wildlife Department. 2024. 2024 Nass River Salmon Stock Assessment - Post-Season Summary. Produced by LGL Ltd. and provided to Pacific Salmon Foundation by Karl English on May 6, 2025.
PSC_20250722	Pacific Salmon Run Size Application. Release 2025.06. Pacific Salmon Commission. [Updated 2025 Jun 26; accessed 2025 Jul 22]. https://www.psc1.shinyapps.io/salmon-run-size-app/
CarrHarris_20250724	Carr-Harris, Charmaine. 2025. Preliminary 2023 estimates of catch and escapement of Skeena sockeye from the Northern Boundary Run Reconstruction Model. Provided to Pacific Salmon Foundation by Charmaine Carr-Harris (DFO) on July 24, 2025.
ProvBC_20240929	Province of British Columbia. 2023. Estimated Escapement of Skeena Summer Steelhead at the Tyee Test Fishery, 1956 - 2024 - All Years to Aug 29, 2024.
Brown_20250703	Brown, Nicholas and Carrie Holt. 2025. Supporting materials and R code for: Brown et al. West Coast of Vancouver Island Natural-Origin Chinook Salmon (Oncorhynchus tshawytscha) Stock Assessment [Data set version 4, accessed Jul 3, 2025.]. Zenodo. https://doi.org/10.5281/zenodo.15801551
Brown_20250417	Brown, Nicholas. 2025. Estimates of spawners, annual terminal run size, harvest, and recruitment for Barkey Sound sockeye Conservation Units. Provided to Pacific Salmon Foundation by Nicholas Brown (DFO) on April 17, 2025.
Jenewin_20240416	Jenewin, Brittany. 2024. Estimates of Fraser river chum salmon catch and escapement from 1998 to 2022 prepared for the Pacific Salmon Commission Chum Technical Committee post-season summary report. Provided to Pacific Salmon Foundatuon by Brittany Jenewin (DFO) on April 16, 2024.
DFO_20250128	Fisheries and Oceans Canada. 2025. Fraser Coho and Chum Post-season Review. Presentation to the Fraser Management Council on January 28, 2025. Available online at https://frasersalmon.ca/wp-content/uploads/2025/01/2024-Fraser-Coho-and-Chum-Post-season-Review-2025-01-28-FINAL-1.pdf [accessed July 3, 2025].
Glavas_20231006	Glavas, Marissa. 2023. Interior Fraser coho abundance and catch time series 1984 - 2022. Provided to Pacific Salmon Foundation by Marissa Glavas (DFO) on October 6, 2023.
DFO_20250225	Fisheries and Oceans Canada. 2025. 2025/26 Southern B.C. Salmon: Chum and Coho. Presentation at the Fraser Approach Forum 2, Richmond B.C., February 25-27, 2025. Available online athttps://frasersalmon.ca/wp-content/uploads/2025/02/2025-Forum-2_Richmond_Chum-and-Coho.pdf [accessed July 3, 2025].
PSC_20250424	PSC Annual Report Application. Release 2025.04. Pacific Salmon Commission. [accessed 2025 Apr 24]. https://www.psc1.shinyapps.io/PSC_Annual_Fraser/
Bison_20241129	Bison, Rob. 2024. Thompson and Chilcotin escapement numbers for steelhead. Provided to Pacific Salmon Foundation by Rob Bison (Provincial steelhead specialist, Ministry of Water, Land and Resource Stewardship, Province of BC) on November 29, 2024.
DFO_20250318	Fisheries and Oceans Canada. 2024. Okanagan Chinook natural-origin and hatchery-origin spawner abundance. Downloaded by Pacific Salmon Foundation from https://github.com/SOLV-Code/Scanner-Data-Processing/blame/main/DATA_IN/Chinook_Okanagan_NEW.csv on March 3, 2025.
Ogden_20250627	Ogden, Athena. 2025. Okanagan sockeye escapement 1961-2024 [data set]. Provided to Pacific Salmon Foundation b y Athena Ogden (DFO) on June 27, 2025.
DFO_20241101	DFO. 2024. Wild Salmon Policy Status, Limit Reference Point, and Candidate Escapement Goals for Okanagan Sockeye Salmon. Canadian Science Advisory Secretariat Science Advisory Report 2024/060. Available from https://publications.gc.ca/collections/collection_2025/mpo-dfo/fs70-6/Fs70-6-2024-060-eng.pdf.

OBMEP_20250228	OBMEP. 2025. 2024 Okanogan Subbasin Steelhead Spawning Abundance and Distribution. Colville Confederated Tribes Fish and Wildlife Department, Nespelem, WA. Report submitted to the Bonneville Power Administration, Project No. 2003-022-00. Available at https://www.okanoganmonitoring.org/ [accessed Mar 18, 2025].
Ogden_20250901	Ogden, A.D., Alex, K., Pestal, G., Alameddine, I., Davis, B., Judson, B., Stiff, H., and Pham, S. 2025. Wild Salmon Policy Status, Limit Reference Point, and Candidate Escapement Goals for Okanagan Sockeye Salmon. Canadian Science Advisory Secretariat Research Document **2025/046**: xii + 102 p. Available from https://waves-vagues.dfo-mpo.gc.ca/library-bibliotheque/41302795.pdf
Parken_20251023	Parken, Chuck. 2025. Fraser Chinook escapement and terminal run by Stock Management Unit, 1979 - 2024. Provided to Pacific Salmon Foundation by Chuck Parken (DFO) on October 23, 2025.

(3) Distribution Information

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(4) Data Quality Information

Data Quality Constraints:
Data quality will vary depending on the source of the raw information. Abundances do not always reflect a total absolute abundance of salmon in the region, and we advise against comparing the absolute abundances across regions and species. In the State of Salmon Report, we present time series as a percent anomaly from the long-term average.

Data Processing:
Data processing is detailed in the following GitHub repository: https://github.com/salmonwatersheds/state-of-salmon. Detailed description of the analytical approach are at: https://salmonwatersheds.github.io/state-of-salmon/.

Process Step 1:

Description: Spawner survey data is read in and used to expand indicator stream counts to regional spawner abundance using the approach in English et al. (2018). Code file 1_regional-expanstions.R (https://github.com/salmonwatersheds/state-of-salmon/blob/main/code/1_regional-expansions.R)

Rationale: Not all regions and species have a regional-scale abundance readily available, so it must be calculated.

Date/time: 2025-04-29

Process Step 2:

Description: Where regional estimates are available, they are read in and compiled into a common format by region and species. When there is no regional estimate available, the output from the expansions (process step 1) are used. Code file 2_compile-regional-data.R (https://github.com/salmonwatersheds/state-of-salmon/blob/main/code/2_compile-regional-data.R).

Date/time: 2024-08-25

Process Step 3:

Description: The trend lines are fitted to the most recent 3 generations and all generations for each time series and predicted abundance added to the dataset. Code file 3_calc-metrics.R (https://github.com/salmonwatersheds/state-of-salmon/blob/main/code/3_calc-metrics.R).

Date/time: 2024-08-25

Process Contact:
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